Accident Analysis and Prevention 190 (2023) 107139

Contents lists available at ScienceDirect

Accident Analysis and Prevention

journal homepage: www.elsevier.com/locate/aap

Passenger and heavy vehicle collisions with pedestrians: Assessment of injury mechanisms and risk

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ARTICLE INFO

Keywords:
Pedestrian
Injury risk
Logistic regression
GIDAS
Heavy vehicles
Passenger vehicles

ABSTRACT

Objective: Automated Driving System (ADS) fleets are currently being deployed in several dense-urban operational design domains within the United States. In these dense-urban areas, pedestrians have historically comprised a significant portion, and sometimes the majority, of injury and fatal collisions. An expanded understanding of the injury risk in collision events involving pedestrians and human-driven vehicles can inform continued ADS development and safety benefits evaluation. There is no current systematic investigation of United States pedestrian collisions, so this study used reconstruction data from the German In-Depth Accident Study (GIDAS) to develop mechanistic injury risk models for pedestrians involved in collisions with vehicles.
Data source: The study queried the GIDAS database for cases from 1999 to 2021 involving passenger vehicle or heavy vehicle collisions with pedestrians.
Methods: We describe the injury patterns and frequencies for passenger vehicle-to-pedestrian and heavy vehicle-to-pedestrian collisions, where heavy vehicles included heavy trucks and buses. Injury risk functions were developed at the AIS2+, 3+, 4+ and 5+ levels for pedestrians involved in frontal collisions with passenger vehicles and separately for frontal collisions with heavy vehicles. Model predictors included mechanistic factors of collision speed, pedestrian age, sex, pedestrian height relative to vehicle bumper height, and vehicle acceleration before impact. Children (≤17 y.o.) and elderly (≥65 y.o.) pedestrians were included. We further conducted weighted and imputed analyses to understand the effects of missing data elements and of weighting towards the overall population of German pedestrian crashes.
Results: We identified 3,112 pedestrians involved in collisions with passenger vehicles, where 2,524 of those collisions were frontal vehicle strikes. Furthermore, we determined 154 pedestrians involved in collisions with heavy vehicles, where 87 of those identified collisions were frontal vehicle strikes. Children were found to be at higher risk of injury compared to young adults, and the highest risk of serious injuries (AIS 3+) existed for the oldest pedestrians in the dataset. Collisions with heavy vehicles were more likely to produce serious (AIS 3+) injuries at low speeds than collisions with passenger vehicles. Injury mechanisms differed between collisions with passenger vehicles and with heavy vehicles. The initial engagement caused 36% of pedestrians' most-severe injuries in passenger vehicle collisions, compared with 23% in heavy vehicles collisions. Conversely, the vehicle underside caused 6% of the most-severe injuries in passenger vehicle collisions and 20% in heavy vehicles collisions.
Significance: U.S. pedestrian fatalities have risen 59% since their recent recorded low in 2009. It is imperative that we understand and describe injury risk so that we can target effective strategies for injury and fatality reduction. This study builds on previous analyses by including the most modern vehicles, including children and elderly pedestrians, incorporating additional mechanistic predictors, broadening the scope of included crashes, and using multiple imputation and weighting to better estimate these effects relative to the entire population of German pedestrian collisions. This study is the first to investigate the risk of injury to pedestrians in collisions with heavy vehicles based on field data.

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https://doi.org/10.1016/j.aap.2023.107139
Received 1 March 2023; Received in revised form 10 May 2023; Accepted 24 May 2023
Available online 13 June 2023
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1. Introduction

By 2020, pedestrian fatalities in the United States (U.S.) had risen 59% from their low in 2009, and in many dense U.S. cities, pedestrians made up a large portion, and sometimes the majority, of fatalities suffered in motor vehicle crashes (NHTSA, 2022). An understanding of collision injury risk enables researchers, designers, and regulators to directly identify the conditions under which real-world injuries occur. Accordingly, injury risk assessments are critical for developing effective strategies to reduce pedestrian injuries and fatalities.

Separately, Automated Driving Systems (ADS) fleets (ADS in this paper is specifically referring to SAE levels 3 through 5 technologies (SAE International, 2021)) are actively being deployed in several dense urban areas in the U.S. In order to safely navigate roadways, ADS are designed with the expectation that they will encounter varying degrees of conflict with other road users, including pedestrians. The ADS design process thus also benefits from being able to estimate injury severity and risk in potential collisions. One example is the usage of injury risk curves in safety benefits analyses of both advanced driver assistance systems (ADAS) and ADS. Employing injury risk curves in the severity evaluation of collisions with and without the equipped technology enables evaluators to determine the injury reduction potential of the technologies ((Rosén et al., 2010); (Scanlon, et al., 2021); (Yanagisawa, Swanson, Azeredo, & Najm, 2017)). This type of comparative analysis extends to collision avoidance testing programs that may be employed to understand the relative safety performance of the ADS with respect to some reference behavior model (Scanlon, Kusano, Engström, & Victor, 2022) (Kusano et al., 2022).

Another example use case for injury risk functions in ADS testing and design is the evaluation of maximum injury potential, whereby a maximum injury risk is computed for a scenario without any collision avoidance action. This maximum injury potential assessment answers the question: how severe could the crash have been if neither actor took collision avoidance action?

Although the deployed ADS fleets in the U.S. are all passenger vehicles, ADS-equipped heavy vehicles are also being tested for future deployment. There is, however, a significant gap in the literature with respect to understanding heavy truck to pedestrian collisions. We were, in our literature review, unable to identify a single reference that specifically examined and modeled heavy truck pedestrian injury risk. This may be due to a number of reasons, including limited regulatory testing of heavy truck crashworthiness, limited field data, and few safety benefits analyses on ADAS/ADS equipped heavy vehicles. This study naturally extends these prior injury risk analyses by asking the research question, how do injury trends and risk change for a heavy vehicle with respect to passenger vehicles?

The ADS use case is important to consider when developing and improving the current state of the art in injury risk functions. There is extensive previous literature estimating injury risk in vehicle to pedestrian impacts; see, for example, (Lubbe et al., 2022), (Niebuhr, Junge, & Rosén, 2016), (Tefft, 2013), (Jin, et al., 2013), (Mallory et al., 2012) and the literature review by (Rosén & Sander, 2009). These studies have traditionally focused on relating vehicle speed or closing speed with injury outcome, and sometimes include pedestrian age. As discussed in (McMurry et al., 2021), the ADS design process also benefits from being able to estimate injury severity and risk in potential collisions. Injury risk models that account for biomechanical and person-specific attributes enable more precise severity evaluation, where the goal is to minimize opportunity for underestimation or overestimation of true risk. With this use case in mind, this study builds on previous work by i) including the most recent data and vehicles; ii) adjusting for additional biomechanically relevant covariates; iii) including children in the modeling; iv) examining injury patterns and risk in collisions with heavy vehicles; and v) conducting sensitivity analyses to understand the effects of missing data and of reweighting towards the entire population of German crashes.

The present study relies on data from the German In-Depth Accident Study (GIDAS) and includes data collected through 2021. Although the study was primarily conceived for U.S. application, to our knowledge, the most recent systematic U.S. reconstruction-based study of vehicle–pedestrian collisions is the National Highway Traffic Safety Administration's (NHTSA's) Pedestrian Crash Data Study (PCDS), which concluded in 1998. Although the German vehicle fleet and driving environment are somewhat different than that of the U.S., we believe that GIDAS represents the most relevant data available in the world today for estimating motor vehicle–pedestrian collision injury risk.

The objectives of this study were to i) describe the injury patterns and frequency for passenger vehicle-to-pedestrian and heavy vehicle-to-pedestrian collision geometries and ii) develop an injury risk function for pedestrians involved in frontal collisions with passenger vehicles and separately with heavy vehicles.

2. Methods

2.1. Data source

We used data from the German In-Depth Accident Study (GIDAS) database for this analysis. GIDAS is a joint project of the German Federal Highway Research Institute (BASt) and the German Association for Research in Automobile Technology (FAT) (German In-Depth Accident Study, 2021). GIDAS has been collecting data on approximately 2,000 crashes annually since 1999 across two survey areas (Dresden and Hanover) and their surroundings. For each crash, investigators at the crash site and hospital collect approximately 3,500 individual datapoints. After documentation, the entire accident sequence is reconstructed in high detail, starting with the accident initiation phase and the reactions of the involved participants, followed by the collision until the final position of the participants. Characteristic variables, such as braking behavior and initial and collision speed of the involved vehicles, are determined. Finally, all data are anonymized.

Since i) the survey areas generalize well to Germany's typical topography, ii) the survey is carried out according to a precisely defined sampling plan, and iii) the number of cases is correspondingly high, the GIDAS database can be used for representative statements about German injury crash occurrence. Given that GIDAS only captures collisions in which at least one person is known to be injured at the time of the crash, reweighting to German national data may be appropriate to account for a slight overrepresentation of serious and fatal collisions (Pfeiffer & Schmidt, 2006).

This study included all data from the GIDAS database as of 31 December 2021, which consisted of 40,686 fully documented and reconstructed crashes (FAT, bast, MHH, VUFO GmbH, 2021). For this analysis of pedestrian crash injury, we included only cases in which a pedestrian experienced a collision with exactly one motorized vehicle. To align with the U.S. categorization of vehicle classes (NHTSA, 2022), our passenger vehicle analysis focused only on collisions with M1 and M2 passenger vehicles weighing less than 10,000 lbs and smaller goods vehicles (N1), according to the official ECE definition (UNECE, 2005) (Table A1).

Our heavy vehicle analysis focused on large vehicles with a flat front. Specifically, it included vehicle classes N2, N3 and M3 by the ECE definition (UNECE, 2005) (Table A1), but excluded N2 vehicles without a flat front, such as vans.

The analysis excluded cases missing the main deformed vehicle area, since we deemed it a mandatory parameter for determining impact direction. Parking or stationary vehicles have additionally been excluded from the dataset, as well as pedestrian collisions with open doors or with trailers. Pedestrians in a wheelchair or pushed in a stroller/buggy have also been excluded due to the different injury kinematics compared with ambulatory pedestrians. Since vehicle speed and pedestrian age are consistently documented strong predictors of injury, cases with unknown values for these variables were also excluded from analysis.

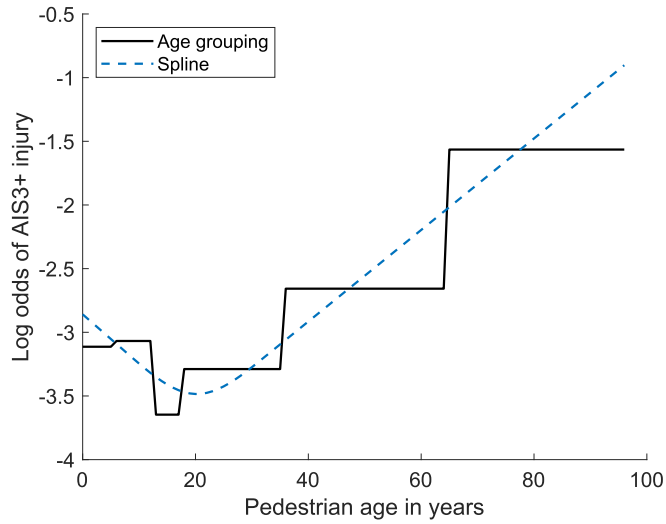

Fig. 2.1. Log odds of MAIS3+ injury in a 20kph collision for age grouping and age spline.

Injury severity was measured using the Maximum Abbreviated Injury Scale (MAIS), which considers the AIS code for the most severe injury sustained in accordance with the 2015 revision of the Abbreviated Injury Scale (AAAM, 2016).

2.2. Injury risk model parameters

Our goal was to develop an injury risk model incorporating parameters associated with known injury mechanisms or presentation of injury. For this reason, our analysis focused on the a priori selected predictors vehicle speed, braking behavior, pedestrian age, sex, and pedestrian size relative to the vehicle; no statistical model selection was performed. Further discussion of these parameters follows.

Vehicle speed is an intuitive and well documented predictor of pedestrian injury risk. It was included as an independent variable for this analysis, with a unique modification. Specifically, we used a parameter term *Impact speed*, defined as the maximum of the vehicle speed at the time of the collision (VK in GIDAS) and the relative speed between the vehicle and the pedestrian (VREL in GIDAS):

$$Impact\ speed = \max(VK(t_{coll}), VREL(t_{coll}))$$

with time of collision t_{coll}. Making this modeling selection represents a conservative approach toward the assessment of pedestrian injury risk.

Older age is consistently associated with increased biomechanical injury risk, but less is known about children's relative injury tolerance. Since children are represented, and in GIDAS over-represented, among pedestrians hit by motor vehicles, we felt it was important they be included. We therefore modeled age using a natural cubic spline in the log odds space with a knot at age 20 and boundary knots at 8 and 32 (R splines package function ns); in practice this contributes two terms (described as *Age Spline 1* and *2* below) to the logistic regression model. We hypothesized that young adults are likely more resistant to injury than younger children and older adults, so we chose a model flexible enough to capture this non-linear relationship should it exist. Note that the spline did not impose this pattern, but it allowed the pattern to be captured if present in the data (Forman & McMurry, 2018). As a sensitivity analysis, we compared the spline model with a model using predefined age categories (0–5, 6–12, 13–17, 18–35, 36–64 and 65+); we found good agreement between the models (see Fig. 2.1) but, in application, continuity and parsimoniousness favor the spline model.

Sex is frequently found to be a significant predictor of biomechanical injury risk (Forman, et al., 2019), and males are significantly over-represented in U.S. pedestrian fatalities (NHTSA, 2022). We therefore included sex as a covariate, with 0 assigned for male and 1 for female.

In a frontal vehicle to pedestrian collision, pedestrian post-collision kinematics are dependent on the nature of the engagement and the pedestrian's pre-impact posture. Most commonly, the collision can result in the pedestrian experiencing a wrap projection or a forward projection. A wrap projection tends to occur when the pedestrian is struck below their center of gravity, such that their upper body wraps around the vehicle front and continues to engage with forward vehicle structures. In a forward projection, the pedestrian is initially projected away from the vehicle and toward the ground, with the opportunity for continued engagement with the vehicle and potential run-over (Wood, Simms, & Walsh, 2005) (Ravani, Brougham, & Mason, 1981). Kinematic outcomes (and hence injury outcomes) can vary considerably and considering the pedestrian's stature relative to the vehicle represents a potential source of this variability. While the height of the leading edge of the front hood of the vehicle is not captured in GIDAS, vehicle bumper height is coded for each vehicle. Additionally, the height of the pedestrian's center of gravity is not encoded in GIDAS, but overall pedestrian height is. As a best-available correlate for capturing this geometric injury mechanism, we included a term measuring the proportion of the pedestrian's height above the bumper:

$$HEIGHTvsBUMP = \frac{Pedestrian\ body\ height - Bumper\ height}{Pedestrian\ body\ height}$$

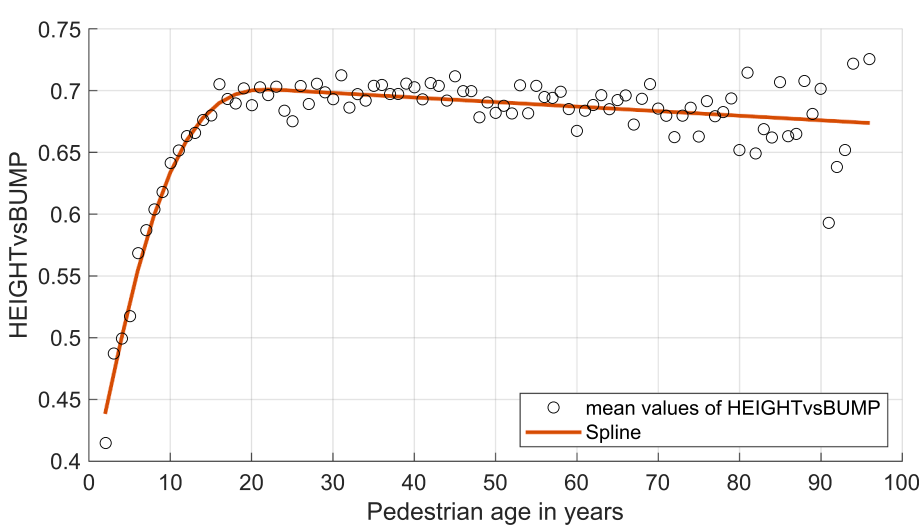

Fig. 2.2. Spline of the ratio HEIGHTvsBUMP depending on the pedestrian age.

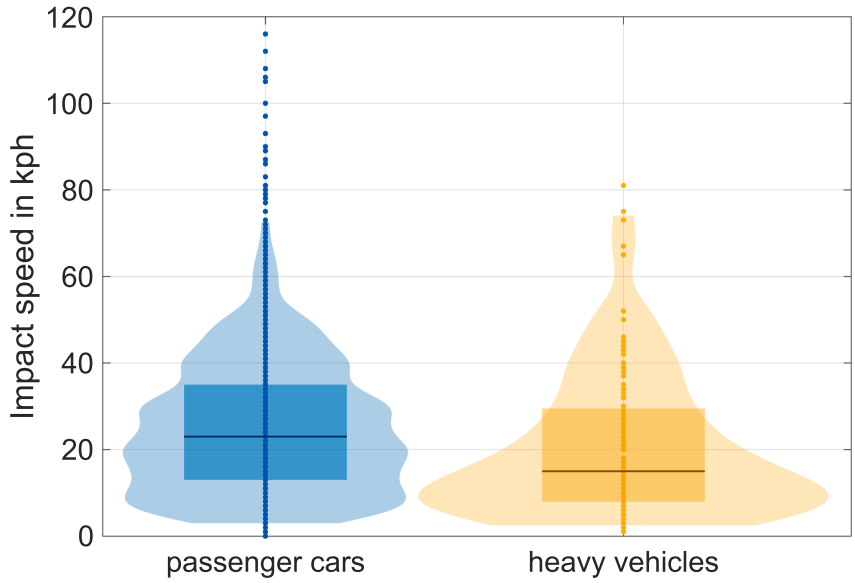

Fig. 3.1. Distribution of the impact speed for all impact directions. Showing the kernel density and a boxplot with the median as solid line and the quartiles.

This value is strongly associated with age for under 20-year-olds, so it needs to be used in conjunction with age to provide reasonable estimates of the injury risk functions; in order to facilitate their use, the mean values of *HEIGHTvsBUMP* in the dataset has been computed for each age and then smoothed using a spline (Fig. 2.2). The values corresponding to this spline are provided in Appendix Table A3.

For heavy vehicles with a flat front, nearly all engagements involve a forward projection. Accordingly, the proportion of the pedestrian's height above the bumper was not considered for the injury risk models involving collisions with heavy vehicles.

Lastly, the extent of continued pedestrian engagement with the vehicle after the initial contact may be related to vehicle braking behavior at the time of collision. We hypothesized that vehicle deceleration at the time of impact would tend to decrease the risk of subsequent pedestrian run-over. We attempted to capture this braking behavior through a binary variable encoded as constant speed/accelerating versus decelerating. The variable *Deceleration* is 0 for accelerating/constant movement and 1 for braking of the vehicle.

Similar to existing work, the injury outcome was modeled using a binary dependent variable associated with MAIS level. For instance, a pedestrian who sustained an injury at the AIS2 level would have their outcome encoded as $Y_i = 1$ when considering the MAIS2+ level of a pedestrian and $Y_i = 0$ for all higher MAIS levels.

We estimated pedestrian injury risk at the MAIS2+, 3+, 4+, and 5+ levels using binary logistic regression, as it is an established method in accident research (Spitzhüttl and Liers, 2016) (Liers, 2018) (Lubbe et al., 2022) with predictors and outcomes as described above. Hence the probability to sustain an injury at one of these levels is modeled by

$$P(Y = 1|X = X_1, \cdots, X_n) = \frac{1}{1 + e^{-z}}$$

where $z = \beta_0 + \sum_{i=1}^{n} \beta_i \bullet X_i$; in the z equation β_0 is the *Intercept*, the β_i are the regression coefficients, and the X_i are the independent variables (I.e. *SEX Female, Impact speed, HEIGHTvsBUMP* (only for passenger vehicles), *Age spline 1, Age spline 2, Deceleration*).

Since injury mechanism is expected to differ by impact direction, and since there were relatively few non-frontal collisions, we developed injury risk functions only for frontal collisions. Collisions were defined as frontal if the initial engagement occurred with the front of the vehicle (GIDAS variable VDI2 = 1) or, in the case of passenger vehicles, with the side of the vehicle forward of the A-pillar. These side collisions with passenger vehicles were also included as frontal collisions because they can result in similar "scooping up" kinematic outcomes for the pedestrian.

2.3. Model evaluation

We evaluated the quality of the models using the Akaike information criterion (AIC), area under the receiver operating characteristic curve (AROC), and area under the Precision-Recall curve (APRC). The ROC curve nicely describes the true-positive-rate and the false-positive-rate for the different classification thresholds. As the passenger vehicle collision dataset is highly unbalanced due to a higher proportion of pedestrians with minor injuries, we also considered the APRC (Saito & Rehmsmeier, 2015).

We analyzed the influence of missing data on the regression through multiple imputation using the R mice package; we used classification and regression trees (cart) (van Buuren & Groothuis-Oudshoorn, 2011) for all variables; and we created ten imputed datasets, each using ten iterations. For imputation, we augmented the dataset with additional variables from the GIDAS dataset, which were expected to provide auxiliary information. For example, to impute pedestrian height, weight is useful in addition to sex and age. Logistic regression models were created for the ten imputed datasets, and the ten estimates for each independent variable were pooled to produce an overall estimate. The models of the original and the imputed master dataset were compared by looking at the regression curves.

We then compared the unweighted models to models where the data was re-weighted to the whole of Germany to eliminate the slight over-representation of serious and fatal collisions in GIDAS. Weighting factors were derived from 2021 German national data (DESTATIS, 2021). The weighting is implemented at the accident level according to three officially defined criteria: the type of accident, the location of accident site, and the injury severity. Since pedestrian collisions are a small subgroup of the overall German crash burden, the Federal Statistical Office of Germany provided us with pedestrian specific distributions. The relative weighting factors were then calculated as:

$$\begin{aligned} &\textit{Relative weighting factor} \\ &= \frac{\textit{Pedestrian Accidents DESTATIS}2021\,(\textit{accident type}, \textit{injury severity}, \textit{location})}{\dfrac{\textit{All Pedestrian Accidents DESTATIS}\,2021}{\textit{All Pedestrian Accidents GIDAS}}} \\ &\bullet \frac{}{\textit{Pedestrian Accidents GIDAS}\,(\textit{accident type}, \textit{injury severity}, \textit{location})} \end{aligned}$$

The normalized weighting factors were defined to be:

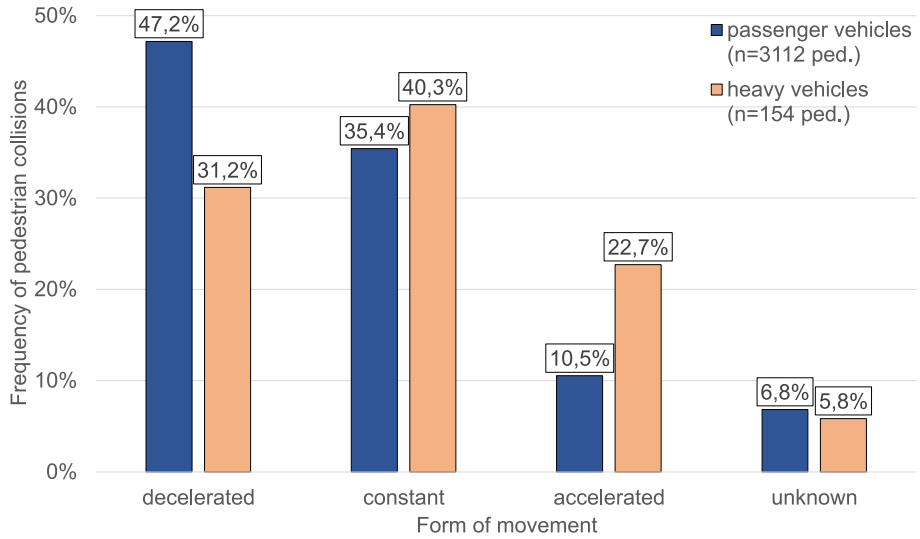

Fig. 3.2. Distribution of the form of movement for pedestrian collisions with passenger vehicles and heavy vehicles.

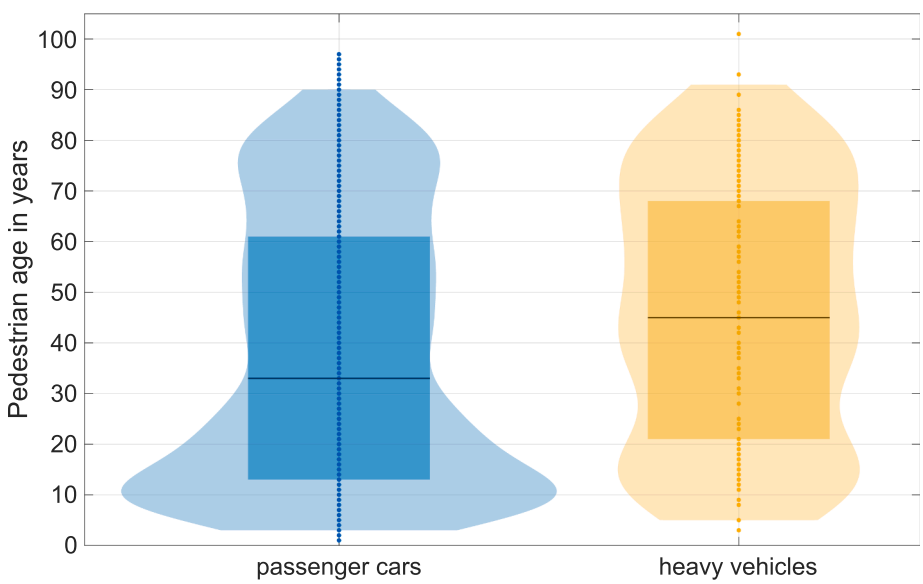

Fig. 3.3. Distribution of the age of the pedestrians for all impact directions. Showing the kernel density and a boxplot with the median as solid line and the quartiles.

$$\textit{Normalized weighting factor} = \textit{Relative weighting factor} \bullet \frac{\textit{no. of pedestrians in dataset}}{\textit{sum (Relative weighting factor)}}$$

Normalized weighting factors create a weighted sample size that equals the size of the unweighted sample (the weights are scaled to have an average value of 1).

Due to identifiability concerns, reweighting was limited to collisions involving exactly one passenger car (M1) and one pedestrian. Weighting factors were not provided for the heavy vehicle collisions; these collisions are relatively rare, leading to concerns about data identifiability. Accordingly, our sensitivity analyses are limited to the passenger vehicle dataset.

Only 21 pedestrians in the heavy vehicle dataset sustained injuries at the AIS4+ level; we therefore restricted our heavy vehicle models to only the MAIS2+ and MAIS3+ levels.

3. Results

3.1. Descriptive analyses

The data sample comprises 3,112 pedestrians involved in collisions with passenger vehicles. A total of 2,524 of these pedestrians were impacted forward of the A-pillar of the vehicle. We further identified 154 pedestrians involved in collisions with heavy vehicles; 87 of them hit by the front of the heavy vehicle. Since the number of pedestrian collisions with heavy vehicles is small, even some factors associated with injury may not show a statistically significant association.

Fig. 3.1 compares the impact velocities for collisions between pedestrians and passenger vehicles and heavy vehicles collisions. Impacts with heavy vehicles tended to be at lower speeds (median: 15kph) than impacts with passenger vehicles (median: 23kph). The impact speed distribution for heavy vehicles shows a peak at 9 kph, a very low speed.

Fig. 3.2, shows acceleration/deceleration across the impacting passenger vehicles and heavy vehicles. Strikingly, 22,7% of the heavy vehicles accelerated prior to the crash while only 31,2% of the heavy vehicles decelerated. In comparison, nearly half (47,2%) of the

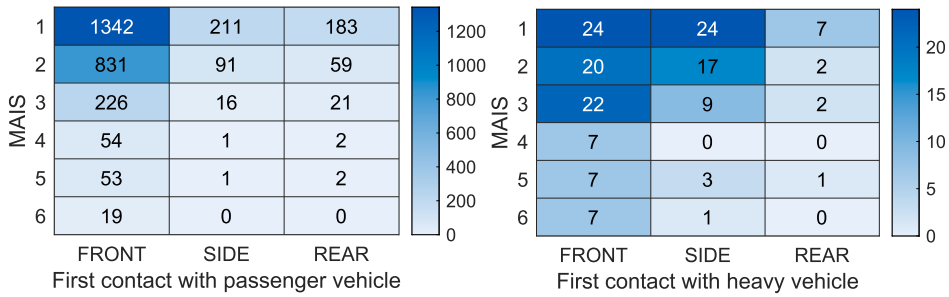

Fig. 3.4. The maximum AIS value of the pedestrian over the first contact with the vehicle (for passenger vehicles: FRONT is defined here as first contact forward of the A-pillar and SIDE is defined as first contact at the vehicle side behind the A-pillar).

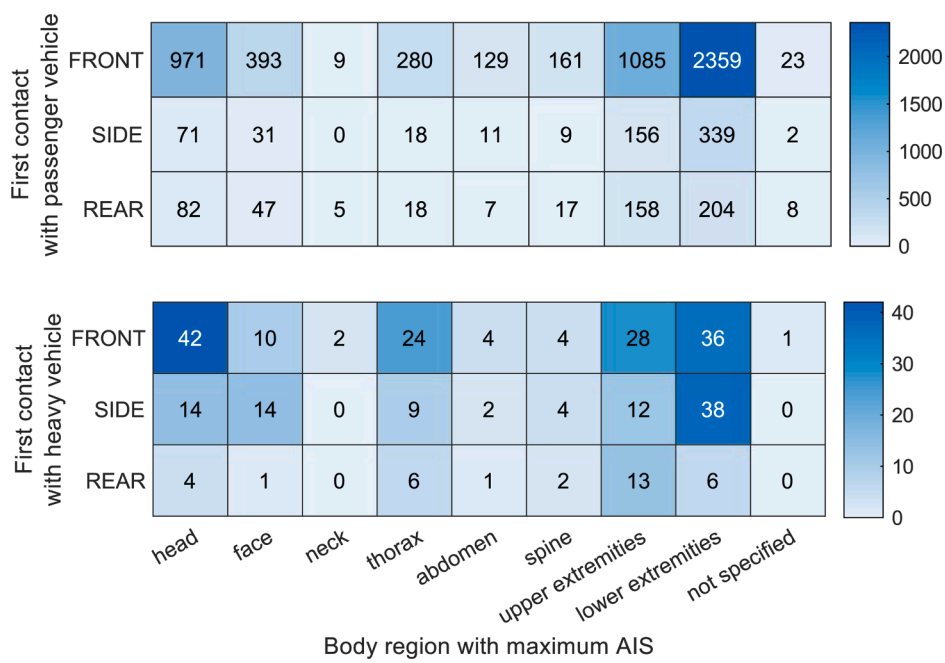

Fig. 3.5. The first contact with the vehicle over injury region with the maximum AIS value of the pedestrian for impacts with a passenger vehicle and heavy vehicle – multiple entries possible (for passenger vehicles: FRONT is defined here as first contact forward of the A-pillar and SIDE is defined as first contact at the vehicle side behind the A-pillar).

passenger vehicle drivers were braking at the time of collision, and only 10,5% were accelerating.

There was also a noticeable difference in the age distributions between pedestrians hit by passenger vehicles and those hit by heavy vehicles (Fig. 3.3). The median age of pedestrians hit by a passenger vehicle was 33, while the median for those hit by a heavy vehicle was 45. The difference was due to a higher proportion of children hit by passenger vehicles, as seen in the peak around 10 years of age.

For both the passenger vehicles and heavy vehicles datasets, approximately 52% of pedestrians struck were female.

3.2. Injury patterns

Even though collisions with heavy vehicles tended toward lower velocities, the injury outcomes were more severe (Fig. 3.4). Approximately 14% of pedestrians involved in a frontal crash with a passenger vehicle suffered AIS3+ injuries compared to 49% of the pedestrians involved in a frontal collision with a heavy vehicle.

The most severely affected body regions are highlighted by heatmap in Fig. 3.5. For each pedestrian, each body region which has an injury at their maximum AIS level is displayed for each impact direction. One person can contribute multiple entries since they could have injuries in different body regions at the same AIS severity. For collisions with heavy vehicles, the most common body regions associated with a pedestrian's highest severity injuries were the head, lower extremities, upper extremities, and thorax. In comparison, for collisions with passenger vehicles, injuries to the lower extremities overwhelmingly represented the most common body region for the highest severity injury. As a point of direct comparison, around 18% of injuries at a pedestrian's highest severity level were to the head in frontal crashes with passenger vehicles, compared with 28% in frontal collisions with heavy vehicles.

Severity levels of the most severe injuries are shown by body region in Fig. 3.6. For both passenger vehicles and heavy vehicles, serious or worse injuries (AIS3+) occurred most often in the head, thorax, and lower extremities. In comparison, the upper and lower extremities account for the majority of injuries at AIS1 and AIS2 levels.

To understand the causes of injury, we examined the injury-causing parts (as coded in GIDAS) for the most severe injuries (Fig. 3.7). In pedestrian crashes with heavy vehicles, the vehicle underside was much more often the injury causing part (19,5%) than in collisions with passenger vehicles (5,8%). For passenger vehicles, the outer vehicle structure (36,1%) and a combination of engagements with the vehicle and/or roadway environment (27,8%) were the most frequent categories.

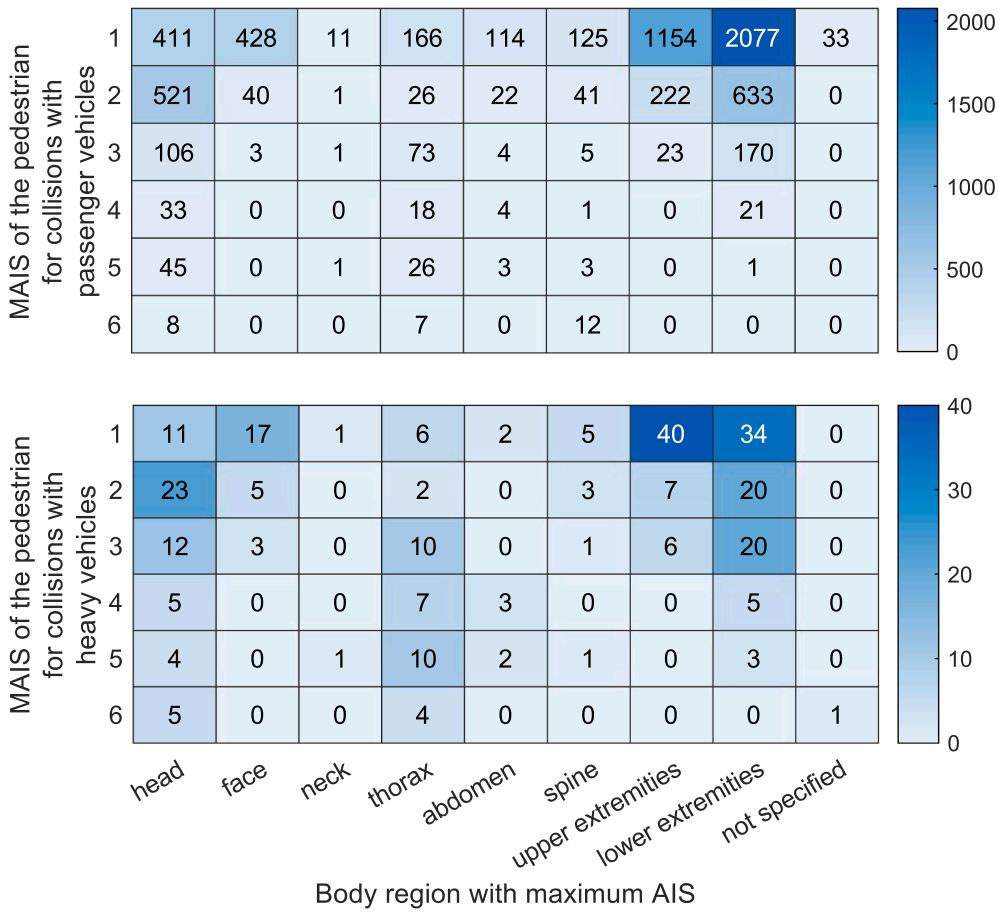

Fig. 3.6. For impacts of the pedestrian with a passenger vehicle or heavy vehicle - the maximum AIS values are displayed over their body region for all injuries with maximum AIS of the pedestrian – multiple entries possible.

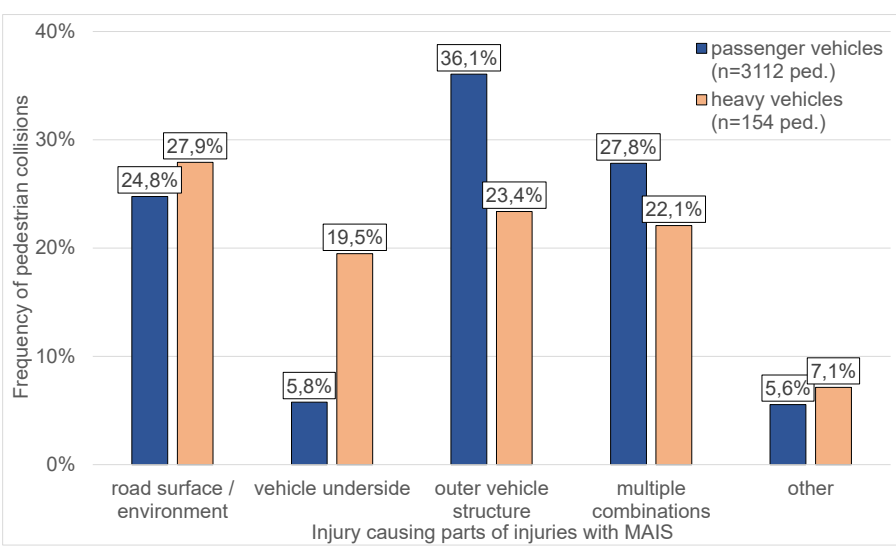

Fig. 3.7. Distribution of injury causing parts for all injuries with the MAIS for passenger vehicles and heavy vehicles for all impact directions.

3.3. Injury risk curves

Due to a low number of cases with MAIS3+ level for side and rear crashes (Fig. 3.4) injury risk functions have only been developed for pedestrian frontal collisions. After excluding unknown values for the independent variables outlined above, the dataset comprised the case numbers in Table 3.1.

As an independent variable, the age spline showed better results when looking at the AIC than the usage of the age groups as an input for the regression models and thus was used in the further analysis. The AROC and APRC are highlighted in the Appendix (Table A2). The APRC values for the MAIS4+ and 5+ models are lower. Due to the larger number of injuries and better model performance, we focus in detail on the MAIS2+ and 3+ models.

Table 3.1
Data sample after excluding unknowns of independent variables, for different injury severities for passenger and heavy vehicles.

		No	*Yes*
MAIS2+	*Passenger vehicles*	867	806
	Heavy vehicles	24	63
MAIS3+	*Passenger vehicles*	1448	225
	Heavy vehicles	44	43
MAIS4+	*Passenger vehicles*	1591	82
	Heavy vehicles	66	21
MAIS5+	*Passenger vehicles*	1626	47
	Heavy vehicles	73	14

Table 3.2
Pedestrian injury risk models in frontal collisions with passenger vehicles.

		Estimate	*Standard Error*	*p-value*
MAIS2+	*Intercept*	−1,459	0,698	0,037
	SEX Female	0,138	0,109	0,205
	Impact speed	0,050	0,004	0,000
	HEIGHTvsBUMP	−0,754	1,152	0,513
	Age spline 1	0,038	0,353	0,000
	Age spline 2	0,265	0,101	
	Deceleration	−0,022	0,109	0,837
MAIS3+	*Intercept*	−1,705	1,083	0,115
	SEX Female	0,022	0,169	0,898
	Impact speed	0,066	0,006	0,000
	HEIGHTvsBUMP	−4,989	1,815	0,006
	Age spline 1	0,884	0,619	0,000
	Age spline 2	0,597	0,180	
	Deceleration	−0,108	0,165	0,514

Table 3.3
Pedestrian injury risk models in frontal collisions with heavy vehicles.

		Estimate	*Standard Error*	*p-value*
MAIS2+	*Intercept*	−1,190	1,078	0,270
	SEX Female	1,124	0,575	0,051
	Impact speed	0,106	0,030	0,001
	Age spline 1	−0,633	0,620	0,217
	Age spline 2	−1,496	2,058	
	Deceleration	−0,231	0,590	0,696
MAIS3+	*Intercept*	−2,198	0,886	0,013
	SEX Female	0,048	0,489	0,922
	Impact speed	0,061	0,019	0,001
	Age spline 1	1,688	1,602	0,086
	Age spline 2	0,666	0,459	
	Deceleration	−0,877	0,547	0,109

The regression coefficients are given in Table 3.2 for passenger vehicle collisions and in Table 3.3 for heavy vehicle collisions for the MAIS2+ and MAIS3+ injury risk curves. Since there are only 5 pedestrians under 18 years with MAIS3+ hit by heavy vehicles, the injury risk function performs poorly under the age of 18 years; this is seen by looking at the logarithm of the odds ratios by age (Fig. A.2).

In some models, the regression coefficients for sex are positive, while in other models they are negative, and in neither case do they reach the traditional $p = 0.05$ threshold for statistical significance. This suggests that after adjusting for height and age, any difference in injury tolerance by sex is at most modest.

The regression coefficients of the impact speed are positive and significant for both passenger and heavy vehicles, indicating increasing injury risk with increasing speed.

An increase in the proportion of the pedestrian's height above bumper height was consistently estimated to predict lower injury risk (all coefficients were negative), and the coefficients in some models reached statistical significance. The effect appeared to be larger in the higher severity injury models (MAIS3+) than in the MAIS2+ models. In other words, for a constant vehicle bumper height, an increase in pedestrian height appears to be associated with decreased injury risk, while, for a constant pedestrian height, an increase in vehicle bumper height appears to be associated with increased injury risk.

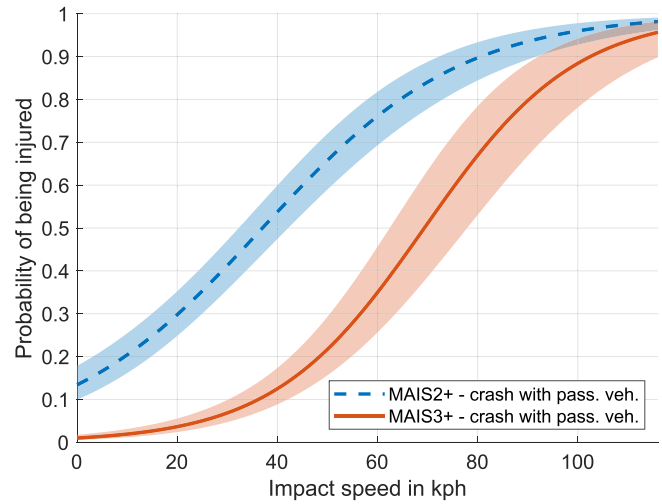

Fig. 3.8. Injury risk of being MAIS2+ or MAIS3+ injured with confidence intervals for passenger vehicles, for AGE = 29 years (median age of pedestrians in passenger vehicle, frontal collisions dataset), SEX = 0, Deceleration = 1, HEIGHTvsBUMP = mean(HEIGHTvsBUMP for 29-year-olds) = 0,69.

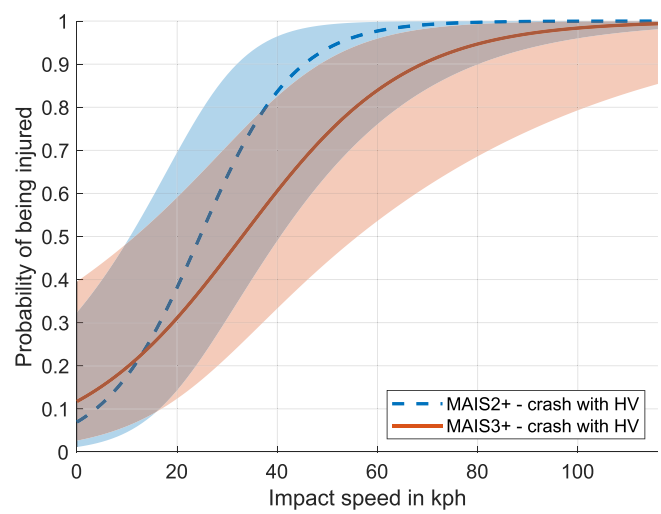

Fig. 3.9. Injury risk of being MAIS2+ or MAIS3+ injured with confidence intervals for heavy vehicles, for AGE = 29 years (median age of pedestrians in passenger vehicle, frontal collisions dataset), SEX = 0, Deceleration = 1.

For the passenger vehicle analysis, the age terms were observed to be statistically significant for all injury risk models. The spline terms are not interpretable on their own, but only in combination. For children, height is also strongly associated with age (Section 2.2), so particularly for pedestrians under age 20, the two spline age terms and the height above bumper term need to be considered together. The regression coefficients for *Deceleration* are almost all negative (with one positive but very close to zero). These coefficients were not statistically significant but might point towards deceleration being associated with decreased injury risk.

The risks of an MAIS2+ or MAIS3+ injury with 95% confidence intervals are shown in Fig. 3.8 and Fig. 3.8. The injury risk curves for passenger vehicles show a clear distinction between MAIS2+ and MAIS3+; the probability for an AIS2+ injury is always higher than for an AIS3+ injury, and the confidence intervals do not overlap. The confidence intervals for heavy vehicles are wide due to the small sample size. For heavy vehicles, the MAIS2+ curve crosses the MAIS3+ curve which,

Table 3.4
Imputed datasets - Pedestrian injury risk models for MAIS3+ in frontal collisions with passenger vehicles.

	Estimate	*Standard Error*	*p-value*
Intercept	−2,665	0,865	0,002
SEX Female	0,138	0,135	0,309
Impact speed	0,070	0,004	0,000
HEIGHTvsBUMP	−3,106	1,432	0,030
Age spline 1	0,124	0,464	0,000
Age spline 2	0,357	0,134	
Deceleration	−0,236	0,137	0,084

since MAIS3+ is a subset of MAIS2+, is not realistic; however, it is expected that with additional data the curves would not cross. For velocities above 13kph, expected injury risk behavior is exhibited, in that the MAIS2+ curve is associated with higher injury risk than the MAIS3+ curve at a given impact speed. Comparing Fig. 3.8 and Fig. 3.8 shows that collisions with heavy vehicles were more likely to produce serious or greater (AIS3+) injuries than collisions with passenger vehicles at a given impact speed.

3.4. Sensitivity analysis

3.4.1. Multiple imputation

Not all cases within GIDAS have complete entries for all fields. We imputed unknown values of the independent variables of sex (0.2% missing), vehicle deceleration (7% missing), and height relative to the bumper (imputed by separately imputing pedestrian height (21% missing) and vehicle bumper height (12% missing)). After 10 iterations, diagnostic plots indicated convergence of the imputation models. The estimated MAIS3+ injury risk model for the imputed dataset is given in Table 3.4. The signs of the estimated coefficients are unchanged from the complete case analysis in Table 3.2.

A comparison between the imputed and complete case injury risk of MAIS3+ for different ages is given in Fig. 3.10. The risk curves are very similar except for smaller children, where a larger number of unknowns existed for pedestrian height and those cases were excluded in the original dataset. In general, children (AGE = 5 years) were at higher risk of injury compared to young adults (AGE = 20 years) due to their age and smaller stature relative to the bumper. After imputation, this risk is even higher than for the original dataset. The highest risk of serious or greater injuries (MAIS3+) existed for the elderly (AGE = 70 years).

3.4.2. Reweighting

Weighting the dataset to the national German statistics gives the estimates for the injury risk model in Table 3.5. In the MAIS3+ model the sign for the regression coefficient of sex is negative, indicating a higher injury risk for male.

The weighted MAIS3+ analysis shows a strong shift towards predicting lower injury risk than the unweighted analysis (Fig. 3.11). This is consistent with serious and fatal crashes being overrepresented in GIDAS.

The MAIS2+ and MAIS3+ injury risk weighted injury risk curves for a 29-year-old (the median age in our data set) are shown in Fig. 3.12.

4. Discussion

This work explores injury patterns and provides updated, mechanistically motivated injury risk functions in passenger vehicle-to-pedestrian and heavy vehicle-to-pedestrian collisions. To our

Table 3.5
Weighted dataset – Pedestrian injury risk models in frontal collisions with passenger vehicles.

		Estimate	*Standard Error*	*p-value*
MAIS2+	*Intercept*	−1,777	0,818	0,030
	SEX Female	0,174	0,124	0,160
	Impact speed	0,040	0,004	0,000
	HEIGHTvsBUMP	−0,880	1,335	0,510
	Age spline 1	0,153	0,409	0,000
	Age spline 2	0,317	0,118	
	Deceleration	0,010	0,121	0,935
MAIS3+	*Intercept*	−2,047	1,590	0,198
	SEX Female	−0,148	0,241	0,540
	Impact speed	0,056	0,006	0,000
	HEIGHTvsBUMP	−4,817	2,636	0,068
	Age spline 1	0,878	0,920	0,000
	Age spline 2	0,622	0,270	
	Deceleration	−0,225	0,230	0,329

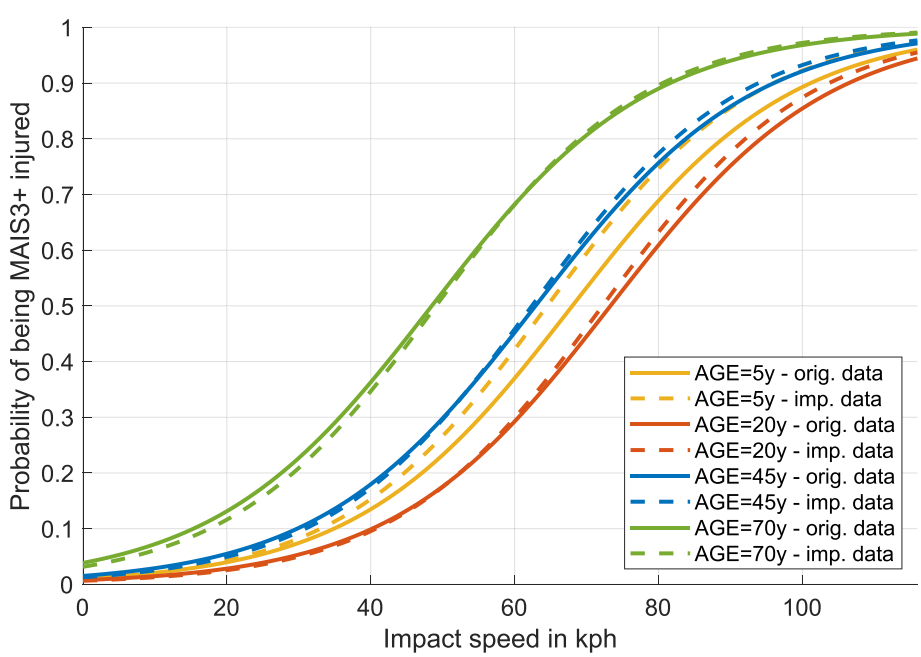

Fig. 3.10. Comparison of MAIS3+ injury risk curves for original and imputed dataset, for different ages, SEX = 0, Deceleration = 1, HEIGHTvsBUMP = mean (HEIGHTvsBUMP for this age).

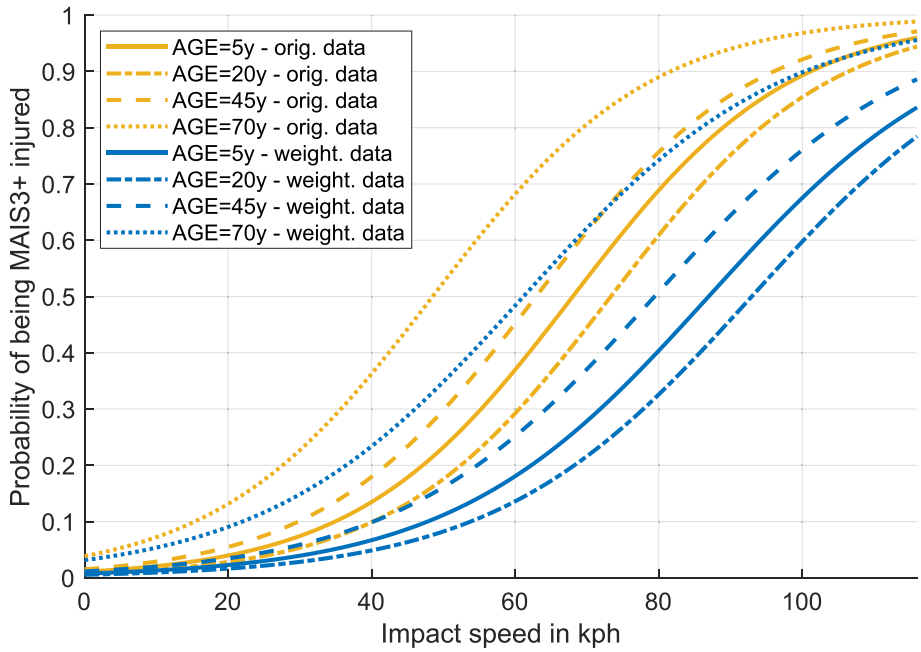

Fig. 3.11. Comparison of MAIS3+ injury risk curves for original and weighted dataset, for different ages, SEX = 0, Deceleration = 1, HEIGHTvsBUMP = mean (HEIGHTvsBUMP for this age).

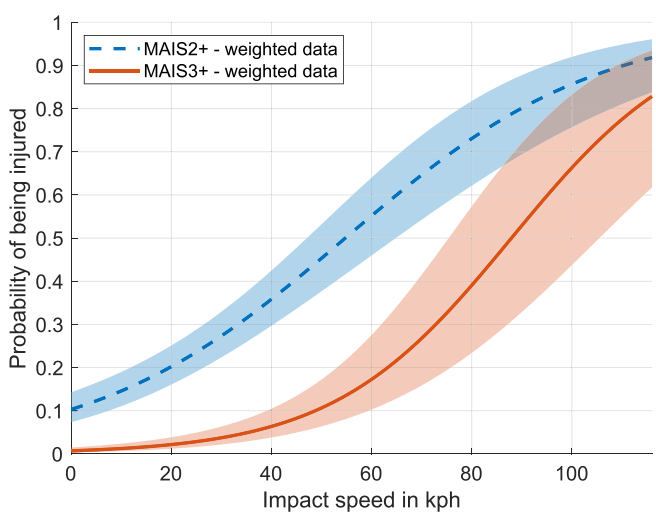

Fig. 3.12. Injury risk curves with weighted data, for pedestrian frontal collisions with a passenger vehicle and AGE = 29 years, SEX = 0, Deceleration = 1, HEIGHTvsBUMP = mean(HEIGHTvsBUMP for 29-year-olds) = 0,69.

knowledge, this is the first field-data-based exploration of injury risk for pedestrians in collisions with heavy vehicles. Our sensitivity analyses suggest that effects on this analysis of missing data in GIDAS are minimal, but that re-weighting towards German national data significantly reduces estimated injury risk.

Our analyses suggest that heavy vehicle collisions have a higher proportion of run-over injuries than passenger vehicle collisions, and that heavy vehicle collisions are higher risk at low speeds than passenger vehicle collisions. The estimates of the models' predictive ability, AROC and APRC, suggest that the heavy vehicle models are less able to discriminate injured from uninjured than the passenger vehicle models. Of note, for heavy vehicles, the MAIS2+ curve crosses the MAIS3+ curve, which may be indicative of vehicle impact speed not being as strong a predictor of injury despite its statistical significance in the model. For pedestrians in a frontal collision with a passenger vehicle, impact speed is the primary factor in injury risk, representing the speed of initial engagement with the vehicle or the speed at which the pedestrian is projected forward of the vehicle. For collisions with heavy vehicles, the risk of pedestrian run-over is present, even at low impact speeds. This may explain why head, thorax, and upper extremities were more frequently injured in heavy vehicle collisions. In a run-over event, the run-over body region and the section of the vehicle underbody engaging with the pedestrian dictate the injury risk rather than impact speed. This difference in the proportion of run-over events between passenger vehicles and heavy vehicles might be due to the poor visibility of pedestrians from the cab of heavy vehicles. As the low speeds suggest, and as seen in deep, single case analyses, the heavy vehicles were often starting up from standstill and failed to see pedestrians walking directly in front of the vehicle.

Our data showed a noticeable peak at around 10 years of age in the distribution of pedestrians hit by passenger vehicles. The age at which German children begin to walk on the streets on their own is around age 6 or 7, and by age 10 a large number of children are doing so. According to (DESTATIS, 2021) 6- to 14-year-olds are particularly likely to have accidents between 7 a.m. and 8 a.m. on Mondays to Fridays, the usual time when children are on their way to school. There is no corresponding peak in collisions with heavy vehicles. It is unclear if this is because children are walking on streets with relatively less exposure to heavy vehicles, because, when compared with cars, heavy vehicles may be more visible from a lower height, or due to another factor.

As with previous pedestrian injury risk models ((Lubbe et al., 2022), (Niebuhr, Junge, & Rosén, 2016), (Tefft, 2013), (Rosén & Sander, 2009)), we observed that age and vehicle speed were significant factors associated with injury risk across all injury severity levels for both passenger vehicle collisions and heavy vehicle collisions. Increased vehicle speed reduces the capacity of a pedestrian to avoid a collision while also increasing the energy imparted to the pedestrian during a collision. Known age-related effects on injury tolerance, reaction time, and postural stability likely explain the consistently significant effect with age. Pedestrian height relative to the vehicle bumper was only observed as a significant factor in the unweighted and imputed MAIS3+ injury risk models, though all models share the association of increased pedestrian height leading to a reduction in injury risk. In general, a taller, upright pedestrian would have their initial contact with a vehicle

at a lower point on their body, further from vital organs, than a shorter pedestrian. The effect of child pedestrians, who are still growing in height, being oversampled in this GIDAS sample also likely plays a role. That the relationship is not significant in the weighted model lends credence to this hypothesis. While we hypothesized that vehicle deceleration at the time of impact may be associated with decreased injury risk due to reducing the potential for continued or secondary engagement with the vehicle following initial contact, this factor was only observed to be statistically significant for the weighted, higher severity passenger vehicle models (MAIS4+ and MAIS5+). It is possible that the principal effect of vehicle braking at the time of impact would be a reduction in impact speed, which is the primary variable associated with injury risk in this dataset, as well as existing injury risk models.

There were inconsistent associations with injury risk by sex, with some models (e.g., weighted MAIS2+) suggesting higher injury risk for females, while other models (e.g., weighted MAIS4+ and MAIS5+) suggested lower risk; this is interesting in the context of the overall rates estimates. While our GIDAS population was close to evenly split among males and females, the most recent crash injury data (2020) indicated that approximately 71% of all pedestrians killed in traffic crashes in the U.S. were males, with a fatality rate more than 2x that observed for females (NHTSA, 2022).

There are several limitations of this study worth noting. As mentioned previously, GIDAS only samples crashes with suspected injury to at least one person. While it is common for police-reported collisions involving a pedestrian to almost always have an associated injury to the pedestrian, there is a bias toward collisions of higher severity in which someone may become injured and a censoring of collisions without injuries. This is exemplified by the injury risk curves having relatively large offsets from 0% risk at impact speeds of 0 kph, especially at the MAIS2+ level. By using data from over 20 years of data collection, older vehicle designs are also captured within the dataset, and the effect of changing vehicle design may not be captured within the developed injury risk models. There are also limitations in the models' ability to fully capture the interaction between the vehicle and pedestrian. For example, neither pedestrian pose/posture nor dynamic changes in vehicle bumper height due to acceleration/deceleration are considered. Further, it can be difficult to differentiate between front and side contacts near the vehicle corner, so any contact area-related effects on risk are not captured within the presented injury risk models. The heavy vehicle models are limited by available sample size, which precluded fitting MAIS4+/5+ models and which further manifests in large standard errors for the regression coefficients in the MAIS2+/3+ models. Finally, differences in the composition of the German vehicle fleet with respect to the current U.S. vehicle fleet may lead to some unquantified accuracy deviations when applying these models for use in the U.S. Despite these limitations, GIDAS represents the best-available source of field crash data.

Our weighting procedure differs from that of (Lubbe et al., 2022). We used the location of the accident as one additional criterion, which seemed important to try to capture similar impact speeds (urban/exurban). Furthermore, we weighted to the distributions of German pedestrian crashes, whereas (Lubbe et al., 2022) used the overall German statistics. And we did not weight cases for each year separately, but rather we weighted the whole dataset to the statistics of 2021 to best reflect the current environment.

Either the unweighted or weighted injury risk functions may be utilized, though care should be made in their application. The unweighted injury risk functions, which are based on the originally sampled GIDAS data, represent a more conservative assessment of injury risk since the database tends to overrepresent higher severity injuries in comparison to the current traffic accident situation in Germany, and therefore predict higher injury risk. While the weighted models should, in theory, better represent the population of German pedestrian collisions, the weights only reflect a small number of factors captured in routine administrative data, and do not explicitly capture important variables such as impact speed. We attempted to account for some of this variability through use of the crash location as an additional weighting factor. Therefore, the weighting towards overall German data may not fully reflect the differences between the GIDAS collision data and the national statistics.

5. Conclusions

This study updates and expands our understanding of the types of injuries that occur in passenger vehicle and heavy vehicle collisions with pedestrians. It further develops injury risk functions incorporating biomechanically relevant predictors and based on the newest available field data.

Funding

Waymo LLC, 1600 Amphitheatre Pkwy, Mountain View, CA 94043, US has commissioned the VUFO – Verkehrsunfallforschung an der TU Dresden GmbH to work on the mentioned research project.

CRediT authorship contribution statement

Angela Schubert: Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing, Visualization. **Stefan Babisch:** Methodology, Software, Formal analysis, Visualization, Writing – review & editing. **John M. Scanlon:** Project administration, Methodology, Writing – review & editing. **Eamon T. Campolettano:** Methodology, Writing – review & editing. **Robby Roessler:** Methodology, Formal analysis, Writing – review & editing. **Thomas Unger:** Project administration, Resources, Writing – review & editing. **Timothy L. McMurry:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Appendix

Table A1
Official vehicle class according to the official ECE definition (Directive 2002/24/EC).

Label	Description
M1	Vehicles used for the carriage of passengers and comprising not more than eight seats in addition to the driver's seat. (Passenger car)
M2	Vehicles used for the carriage of passengers, comprising more than eight seats in addition to the driver's seat, and having a maximum mass not exceeding 5 tonnes
M3	Vehicles used for the carriage of passengers, comprising more than eight seats in addition to the driver's seat, and having a maximum mass exceeding 5 tonnes
N1	Vehicles used for the carriage of goods and having a maximum mass not exceeding 3.5 tonnes.
N2	Vehicles used for the carriage of goods and having a maximum mass exceeding 3.5 tonnes but not exceeding 12 tonnes
N3	Vehicles used for the carriage of goods and having a maximum mass exceeding 12 tonnes.

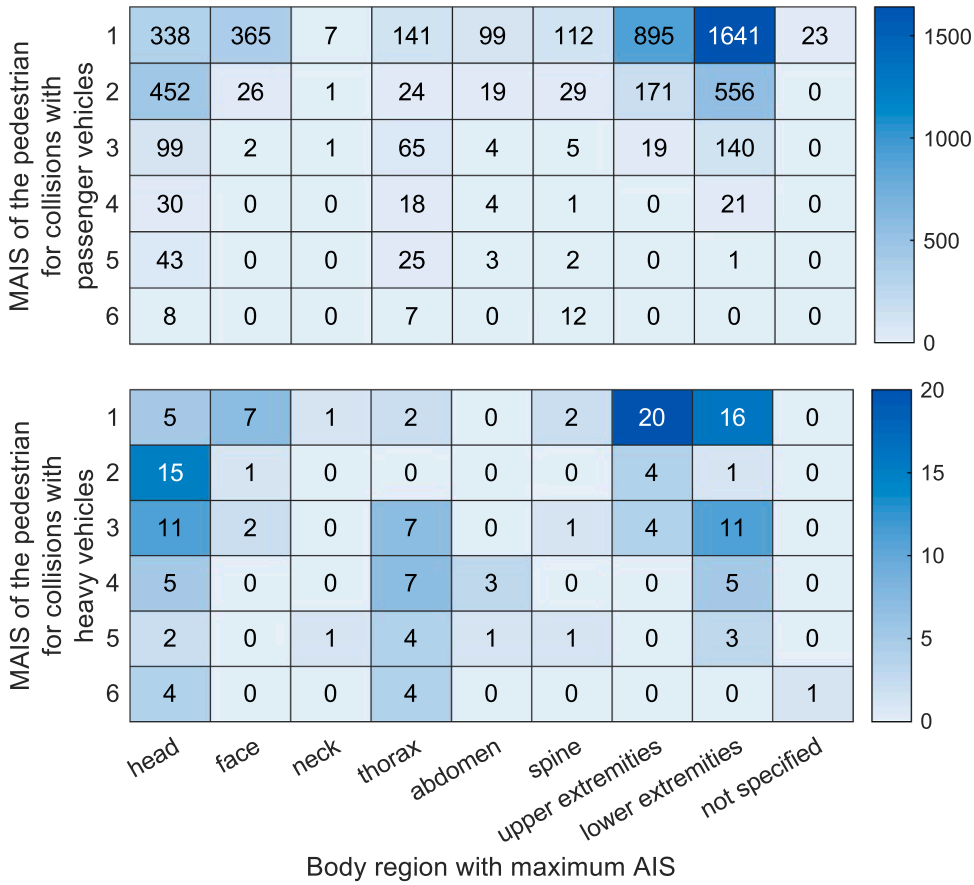

Fig. A1. Distribution of severity by body region for injuries with severity equal to each pedestrian's MAIS (multiple entries per pedestrian are possible if the pedestrian had the same severity injuries to multiple body regions). Frontal collisions only.

Table A2
Area under ROC and PRC curve for injury risk models.

		passenger vehicles	*heavy vehicles*
MAIS2+	*AROC*	0,72	0,76
	APRC	0,71	0,87
MAIS3+	*AROC*	0,81	0,64
	APRC	0,43	0,63
MAIS4+	*AROC*	0,88	
	APRC	0,34	
MAIS5+	*AROC*	0,87	
	APRC	0,22	

Table A3
The independent variable HEIGHTvsBUMP depending on age.

Age in years	*HEIGHTvsBUMP*
2	*0,438*
4	*0,499*
6	*0,554*
8	*0,599*
10	*0,634*
12	*0,659*
14	*0,678*
16	*0,690*
18	*0,697*
20	*0,700*
22	*0,701*
24	*0,700*
26	*0,700*
28	*0,699*
30	*0,698*
32	*0,697*
34	*0,697*
36	*0,696*
38	*0,695*
40	*0,694*
42	*0,694*
44	*0,693*
46	*0,692*
48	*0,691*
50	*0,691*
52	*0,690*
54	*0,689*
56	*0,689*
58	*0,688*
60	*0,687*
62	*0,686*
64	*0,686*
66	*0,685*
68	*0,684*
70	*0,683*
72	*0,683*
74	*0,682*
76	*0,681*
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82	*0,679*
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96	*0,674*

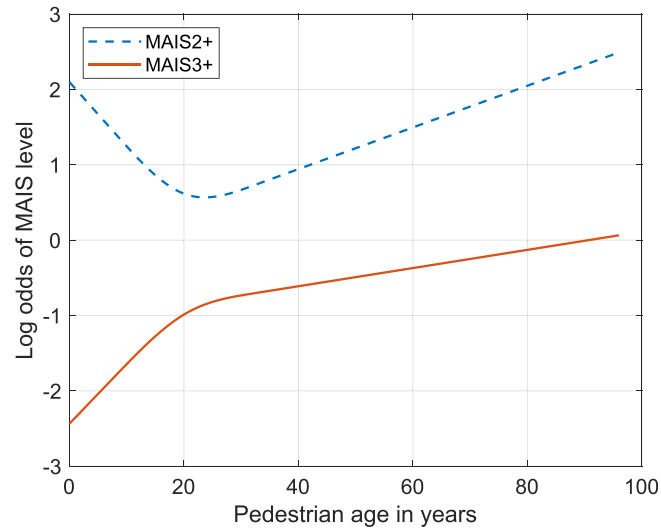

Fig. A2. Log odds ratio $\log\left(\frac{p}{1-p}\right)$ of MAIS2+/3+ depending on age for impact speed = 20kph, SEX = female, Deceleration = 1 for the injury risk functions of heavy vehicle collisions from Table 3.3.

Table A4
Unweighted MAIS 4+ and MAIS5+ pedestrian injury risk models in frontal collisions with passenger vehicles.

		Estimate	*Standard Error*	*p-value*
MAIS4+	*Intercept*	−4,724	2,092	0,024
	SEX Female	−0,459	0,313	0,143
	Impact speed	0,070	0,008	0,000
	HEIGHTvsBUMP	−2,155	3,456	0,533
	Age spline 1	−0,245	1,240	0,000
	Age spline 2	0,375	0,362	
	Deceleration	−0,451	0,297	0,128
MAIS5+	*Intercept*	−4,247	2,756	0,123
	SEX Female	−0,719	0,427	0,092
	Impact speed	0,069	0,010	0,000
	HEIGHTvsBUMP	−3,836	4,629	0,407
	Age spline 1	0,331	1,676	0,000
	Age spline 2	0,502	0,493	
	Deceleration	−0,692	0,401	0,084

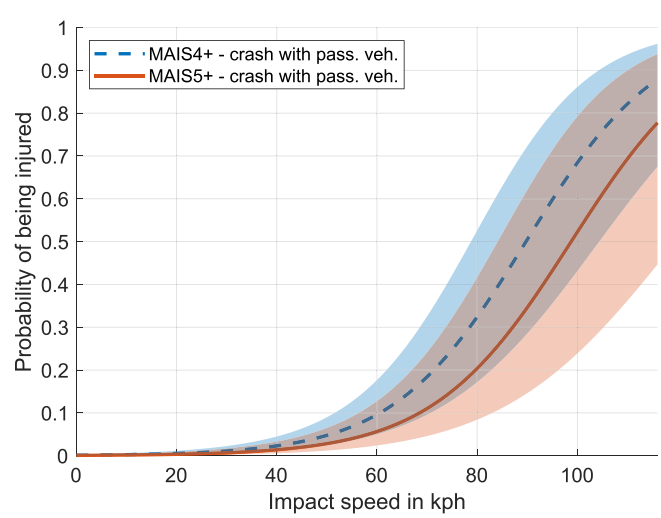

Fig. A3. MAIS4+ and MAIS5+ injury risk curves and confidence intervals for a 29-year-old pedestrian hit by a passenger vehicles, SEX = 0, Deceleration = 1, HEIGHTvsBUMP = mean(HEIGHTvsBUMP for 29-year-olds) = 0,69.

Table A5
Weighted dataset – Pedestrian injury risk models in frontal collisions with passenger vehicles.

		Estimate	*Standard Error*	*p-value*
MAIS4+	*Intercept*	−4,591	2,757	0,096
	SEX Female	−0,888	0,419	0,034
	Impact speed	0,054	0,008	0,000
	HEIGHTvsBUMP	−1,912	4,569	0,676
	Age spline 1	−0,563	1,675	0,000
	Age spline 2	0,334	0,494	
	Deceleration	−0,742	0,393	0,059
MAIS5+	*Intercept*	−5,310	3,595	0,140
	SEX Female	−1,186	0,557	0,033
	Impact speed	0,056	0,009	0,000
	HEIGHTvsBUMP	−1,605	5,970	0,788
	Age spline 1	−0,641	2,210	0,000
	Age spline 2	0,330	0,656	
	Deceleration	−1,044	0,520	0,044

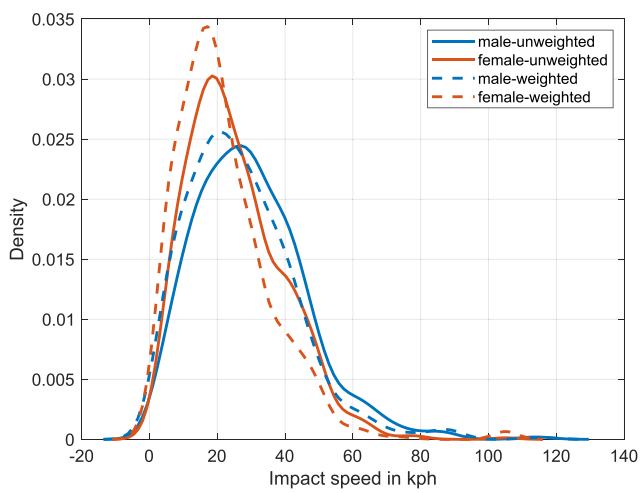

Fig. A4. Kernel density of the impact speed for weighted and unweighted dataset - female/male.

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